

Continuity, Extended: the frontier build and the objection kill-sheet

Research into making the product substantially more ambitious: what real prior art makes possible, which mechanisms defuse each objection, and what stays in the weekend build versus the vision slides.

Prepared 22 August 2026 · Companion to the Strategy Continuity paper and the main Research Dossier · New citation prefixes: **C** capture and friction, **I** incentives and adoption, **G** AI frontier and risk quantification, **L** legal and regulatory. All sources in Section 7. Everything here is independent research; nothing is from the track brief.

1 · What this addendum found, in one paragraph

Three discoveries reshape the pitch. First, the friction objection is solvable with published patterns: provenance tools that capture with literally zero workflow change exist as open source [C1-C6], and the research shows the real bottleneck was never writing effort but deciding what to record [C19], which is exactly what an LLM agent automates. Second, regulators have already written our product spec: the FCA criticized firms for taking "key decisions... without sufficient explanation of the underlying rationale" [L7], the Fed defines model documentation as detailed enough that "parties unfamiliar with a model" could understand it [L9], and UK handover rules literally require material containing "judgement and opinion, not just facts and figures" [I17]. Third, the strongest new fact in the whole sweep is legal: US courts dismiss trade-secret cases, regardless of whether theft actually happened, when the firm cannot describe its secret with "reasonable particularity" [L2], and roughly 11% of cases die on the "reasonable measures" requirement alone [L1]. An undocumented strategy may be legally undefendable. That upgrades Continuity from a nice-to-have into the thing that determines whether the Jane Street lawsuit is even possible.

2 · The upgraded architecture

Version 2 of the product is five layers. The weekend build implements slices of layers 1, 3 and 5 (Section 6); the rest is the vision, now grounded item by item.

LAYER	WHAT IT DOES	GROUNDING
1 Capture	Zero-friction provenance: every run, diff and parameter change recorded as a side effect of working; LLM drafts the rationale, the human approves	noWorkflow, Verdant, papermill pattern [C1, C2, C5]; Copilot PR summaries [C8]
2 Firm brain	A knowledge graph over the firm's exhaust (code, chat, notebooks) answering global questions with citations	Microsoft GraphRAG [G1]; rationale-mining field [G3, G4]
3 Elicitation	The debrief agent: LLM interviewer that detects significant decisions, asks targeted questions at natural pauses, schedules follow-ups before memory decays	LLMREI [G11]; Duolingo half-life regression [G13]; Morgan Stanley Debrief [G14]
4 Continuity assets	Handover packs, onboarding courseware, warm-start priors for successors, and (the headline) the researcher twin	SYSC 25.9 handover spec [I17]; AutoElicit [G12]; Stanford generative agents [G6]

5 Proof and risk	Cryptographic timestamping and tamper-evident logs; knowledge-at-risk metrics (bus factor, vacation-readiness) and auto-drafted compliance artifacts	RFC 3161, Merkle logs [G17, G18]; truck-factor algorithm [G21]; RTS 6, SR 11-7 [L6, L9]
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3 · The frontier capabilities, each with its honest limit

3.1 Zero-friction capture (kills the friction objection)

The academic record shows capture can be invisible. noWorkflow collects full provenance of Python research code requiring "neither changes to the script, nor any laborious configuration" [C1]; Verdant (a CMU best-paper) records every notebook cell and output edit passively in the background [C2]; nbsafety tracks cell-level lineage from a drop-in kernel, and the cells it flagged were seven times more likely to be what users actually re-ran [C4]; Netflix made every notebook run self-documenting by construction with papermill's immutable output artifacts [C5]; and JupyterLab ships an official telemetry extension point, so none of this requires forking the tools quants already use [C6]. On the writing side, LLMs already beat state-of-the-art at generating the "what changed and why" record from a diff [C7], and GitHub ships PR summaries commercially, with a documented rule we adopt: the model drafts, the human approves, because hallucinated rationale is worse than none [C8]. The design also respects an empirical constraint: developers switch activities every 0.3 to 2.0 minutes [C15], so Continuity never interrupts mid-flow; it batches approvals at natural boundaries (merge, promotion to paper trading), which is where the ADR field-study says documentation actually succeeds [C18, C19]. *Limit: capture of artifacts is total; capture of reasoning still depends on the human approving or correcting drafts.*

3.2 The firm brain (kills the "just a wiki" objection at scale)

Microsoft's GraphRAG builds an entity knowledge graph from source documents and pre-summarizes communities of related entities, enabling global questions like "what do we collectively know about crowding in momentum?" answered with citations [G1]; it is open source. Repository-level code question-answering is now a benchmarked research task (576-question SWE-QA spanning "intention understanding, cross-file reasoning, and multi-hop dependency analysis") [G2], and mining decision rationale from commit histories is an established research field with current tooling (CoMRAT detects decision and rationale sentences in commit messages) [G3, G4]. The who-knows-what layer has twenty years of precedent: Expertise Browser computed expertise maps from change history at Bell Labs in 2002 [G24]. *Limit: graph indexing is LLM-expensive, quant jargon needs a custom extraction schema, and hallucinated graph edges must be provenance-checked against the underlying artifacts.*

3.3 Active elicitation, scheduled like spaced repetition

The 1980s named this exact problem: the "knowledge acquisition bottleneck" was the blocker that killed expert systems [G10]; the pitch line is that LLMs are the first credible attack on a forty-year-old bottleneck. Evidence they work as interviewers: an LLM requirements-elicitation interviewer made a similar number of errors to human interviewers and generated contextually appropriate follow-ups [G11]. Morgan Stanley already ships an AI meeting "Debrief" agent to its advisors, with adoption above 98% of advisor teams for its knowledge assistant [G14], proving the pattern lands in regulated finance. Our twist: schedule the questions with half-life regression (the Duolingo spaced-repetition model [G13]), so the agent asks about a decision just before organizational memory of it decays, and only when decision-detection heuristics fire (a threshold changed after a drawdown, a branch abandoned after N backtests). *Limit: elicitation*

quality in quant research specifically is unstudied, and interviewees can satisfice; mitigated by grounding every question in a concrete captured artifact.

3.4 The researcher twin (the headline, stated honestly)

Stanford and DeepMind built generative agents of 1,052 real people from a two-hour structured interview each; the agents answered social-survey questions at about 85% of the accuracy of the person's own two-week test-retest consistency [G6]. Applied here: a departing researcher's exit debrief plus their captured decision corpus becomes a queryable agent; the successor asks "why did you cap position size in the expiry window?" and gets the answer the departed quant would have given, linked to the records that support it. Commercial precedent exists (Delphi's "digital minds") [G9], and captured beliefs can do more than answer questions: AutoElicit shows elicited expert knowledge can be converted into Bayesian priors that warm-start the successor's models with fewer labels [G12]. State the limits on stage, because they are provable: imitation-learning theory shows clone error compounds off the expert's beaten path [G7], and Polanyi's tacit-knowledge problem ("we know more than we can tell") bounds what any corpus contains. The twin is a memory prosthesis for the desk, not a replacement trader; a finance-specific cloning attempt exists and regime-shift fragility is its known weakness [G8]. If the firm fine-tunes twin models, instructional fingerprinting lets it prove model ownership even after further fine-tuning [G20].

3.5 The cryptographic proof layer (turns the record into a legal weapon)

Every captured decision gets an RFC 3161 trusted timestamp (the IETF standard, with ANSI X9.95 as its financial-grade extension) [G17], chained into a Merkle-tree tamper-evident log of the kind that runs Certificate Transparency [G18]; OpenTimestamps provides free git-integrated anchoring implementable in an afternoon [G19]. Why this matters legally: contemporaneous records "kept in the course of a regularly conducted activity" are exactly what the business-records exception to hearsay admits into evidence [L20]; US broker-dealer rules already require records in tamper-evident form with a designated third party able to hand them to regulators [L19], which is direct precedent for "neutral third-party record" being a feature rather than a bug; and the DTSA case law (next section) makes a precisely-documented secret the difference between winning and being dismissed. *Limit: timestamps prove when and what, never who truly originated an idea; courts weigh them as strong supporting evidence, not a silver bullet.*

3.6 Knowledge risk, quantified like the firm quantifies everything else

The bus factor is a real algorithm, not a metaphor: published methods compute it from git history (46% of popular GitHub projects have truck factor 1) with an open-source tool to reuse [G21]; a practitioner survey found 75% rate it important, 63% hit it in the last year, only 19% ever saw it measured, and one respondent reported six person-months recovering from a single departure [I19]. Turnover-induced knowledge loss has been modeled as a rare-severe-event risk (the framing lets us pitch "knowledge VaR" to a risk-fluent audience) [G22], and Microsoft's ownership research shows concentration measurably predicts defects [G23]. The regulatory hook is delightful: finance already mandates a bus-factor fire drill. FINRA guidance and NY DFS letters require traders to take about two consecutive weeks of leave with access cut off, precisely because misconduct "requires the continued presence of the wrongdoer" [I18]. Continuity ships a **vacation-readiness score**: whether the desk passes its mandated two-week absence cleanly. *Limit: authorship metrics count who touched things, not who could rebuild them; the debrief corpus is what closes that gap.*

4 · The objection kill-sheet

Each objection from the Strategy Continuity paper, now answered with researched mechanisms rather than assertions. This table is built to survive judge Q&A.

OBJECTION	THE RESEARCHED ANSWER
"Traders will refuse to feed it" (hoarding is rational)	Four stacked mechanisms. (1) Pay for it: Marshall Wace's TOPS paid contributors by measured idea performance (over \$250M/yr by 2004-05) and grew to ~\$30B of AUM; WorldQuant pays external consultants per alpha, top tiers earning \$8,000+ quarterly [I6, I7, I8]. Continuity's decision records become P&L-attributed "idea tickets" wired to bonus pools. (2) Make the record the trader's asset: employers demonstrably trust third-party-verified contribution traces ("GitHub is your new resume") [I13], so a firm-signed decision record is comp-negotiation ammunition; documentation becomes leverage, not its loss. (3) The research says fear of losing knowledge-power is empirically weaker than assumed, intrinsic motivators and visible credit dominate (badge introductions raised contribution by a median 88.6%, strongest among senior users) [I1, I11, I12], and hoarding measurably backfires on the hoarder via distrust loops that reduce their own creativity [I3]. (4) The dilemma has a known game-theoretic fix: restructure payoffs, show contributors their contribution mattered, make group identity salient [I2]. Design rule from Bridgewater's cautionary tale: capture decisions, never rank-and-shame people; their recorded-everything culture worked as capture but the published league tables produced fear and churn [I9, I10].
"It's surveillance"	Finance is the one industry fined for NOT recording: \$3B+ against 100+ firms since 2021 for failing to capture business communications [L12]; UK rules mandate recording even intent-stage conversations, retained five to seven years [L13]; trader voice transcription is installed infrastructure [C12]. Continuity structures what is already legally captured, and adopts the Microsoft Recall lesson for the boundary: opt-in, artifact-level (never screens), encrypted, transparent [C14]. In the EU the boundary is law, not taste: German works councils have co-determination over monitoring tech and a DPIA is required [L15, L14], so the product is architected as decision-record capture of work product, not behavioral analytics. The trader-side sweetener: WakaTime proves developers voluntarily run always-on telemetry when it gives them personal value back [C17]; every researcher gets their own searchable history first.
"Tacit knowledge can't be captured"	Conceded, and now bounded instead of hand-waved: imitation theory proves clones degrade off-distribution [G7], Polanyi's limit is real. But the addressable layer is large and the marginal value is enormous: the Stanford result shows a two-hour interview captures enough to hit ~85% of a person's own consistency on held-out questions [G6], elicited knowledge can warm-start successor models as priors [G12], and the legal system only requires the explicit layer anyway: trade-secret cases are won on documented particularity, not on tacit essence [L2].
"Firms would build it in-house"	For compliance-grade records, the world's most sophisticated banks overwhelmingly buy: Global Relay serves 22 of the top 25 banks, Smarsh 18 of the top 20 [L18]. The reason generalizes to us: US securities law itself requires a designated third party able to deliver records to regulators "even if the firm is unwilling or unable" [L19], and neutral, routinely-generated records carry evidentiary weight internal reconstructions lack [L20]. Continuity is the same buy logic one layer up: reasoning instead of messages.
"Just a wiki with hooks"	Wikis fail at the selection step, not the writing step [C19]; Continuity automates selection (decision detection), drafting (LLM from diffs [C7]), recall (graph queries with citations [G1]), scheduling (half-life regression [G13]), proof (timestamped tamper-evident log [G17, G18]), and measurement (truck factor, knowledge VaR [G21, G22]). No wiki does any of those six.

"Who is the buyer?"	Three, all above the trading floor. Legal: the general counsel, because ~11% of trade-secret cases die on reasonable measures and courts demand reasonable particularity the firm may not possess [L1, L2]; attorney commentary on Jane Street v. Millennium literally prescribes "document development and implementation meticulously" [L4]. Compliance: RTS 6 Article 5(7) requires a who/what/when/approved-by record of every material algo change, and SR 11-7 defines documentation as understandable by "parties unfamiliar with a model" [L6, L9]; the FCA has already cited firms for missing rationale [L7]. Capital: ILPA standards make a key person's departure an automatic suspension of the investment period, allocator DDQs demand documented succession plans, and half of hedge fund failures are operational, not investment, failures [L16, L14, L17]. The COO buys continuity, the GC buys evidence, IR buys DDQ answers; the trading floor just has to tolerate it, which is what Section 4 row one is for.
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5 · The three "crazy but real" demo peaks

For the video and live pitch, in escalating order of audacity, each fully backed:

1. **The vacation-readiness score.** Show a desk failing its regulator-mandated two-week absence test before it happens [I18, G21]. One number, instantly legible to any finance judge, and no other team will have found the block-leave rules.
2. **The mock lawsuit.** Replay the Jane Street scenario against the synthetic firm: the departing trader's counsel claims "it was just his experience and expertise"; Continuity produces the timestamped, tamper-evident decision genealogy with reasonable particularity, and the slide cites the case law where its absence gets claims dismissed [L2, L1, G17]. This lands the triple-value framing in thirty seconds.
3. **Interviewing the departed quant.** The researcher twin answers the successor's question, citing its sources from the corpus, followed immediately by the honesty slide: what the twin can and cannot know, with the imitation-learning bound on screen [G6, G7]. Judges reward the audacity and remember the honesty.

6 · Weekend triage: what gets built vs pitched

CAPABILITY	WEEKEND VERDICT	NOTE
Capture hooks + LLM-drafted decision records (approve/edit flow)	Build	Core loop; all components proven [C1, C7, C8]
Bus-factor dashboard + departure simulation + vacation-readiness score	Build	Reuse the open-source truck-factor algorithm [G21]; the score is arithmetic on top
Debrief agent (session-end questions grounded in captured diffs)	Build	An LLM prompt over the capture stream; half-life scheduling can be stubbed [G11, G13]
OpenTimestamps anchoring of the decision log	Build (half a day)	Free, git-integrated; makes the mock-lawsuit demo real [G19]
Handover pack generator	Build	Template it on the SYSC 25.9 spec: "judgement and opinion, not just facts and figures" [I17]
Researcher twin	Demo-slice only	A RAG agent over the synthetic corpus in the departed persona is honest and impressive; do not claim fine-tuning [G6]
GraphRAG firm brain	Stretch goal	Open-source pipeline exists but indexing cost eats time; a small pre-built graph over the synthetic corpus is enough for one query on stage [G1]

Auto-drafted RTS 6 / SR 11-7 artifacts	One sample page	Generate a single filled self-assessment section from the capture stream; the citation does the rest [L6, L9]
Warm-start priors, model fingerprinting, comp integration	Pitch only	Vision slides with citations [G12, G20, I6]; claiming them as built would be dishonest and judges will probe

7 · Citations

Compact list; codes as used above. Quotes shown in Sections 1-6 are verbatim from fetched pages unless the entry notes otherwise. Items from paywalled or abstract-only sources are marked; verify before putting numbers on slides.

C · Capture and friction

- C1** · noWorkflow (VLDB 2017 + OSS): zero-config provenance for Python research code. <https://github.com/gems-uff/noworkflow>
- C2** · Verdant, CMU (VL/HCC 2018 best paper): passive background history of notebook experimentation. <https://marybethkery.com/Verdant/>
- C3** · ProvBook: per-cell execution provenance with diff UI. <https://ceur-ws.org/Vol-2180/paper-57.pdf>
- C4** · nbsafety (VLDB): drop-in kernel lineage; flagged cells 7x more likely to be re-executed. <https://arxiv.org/abs/2012.06981>
- C5** · Netflix papermill infrastructure: every notebook run emits an immutable, self-documenting output artifact. <https://netflixtechblog.com/scheduling-notebooks-at-netflix-348e6c14cfd6>
- C6** · JupyterLab telemetry (official): schema-based event hooks without forking. <https://github.com/jupyterlab/jupyterlab-telemetry>
- C7** · LLM commit-message generation study: superior to prior automated methods. <https://arxiv.org/abs/2404.14824>
- C8** · GitHub Copilot PR summaries + hallucination warning ("reviewing is a requirement"). <https://docs.github.com/en/copilot/responsible-use/pull-request-summaries>
- C12** · NICE Actimize trader voice transcription; Verint Capture: comms capture is installed infrastructure at trading firms. <https://www.niceactimize.com/financial-markets-compliance/trader-voice-transcription>
- C14** · Microsoft Recall backlash and redesign: off by default, authenticated, encrypted; the acceptability blueprint. <https://www.notebookcheck.net/Microsoft-makes-three-critical-changes-to-Windows-Recall-after-facing-backlash.845469.0.html>
- C15** · Meyer et al., IEEE TSE: developers stay 0.3-2.0 minutes per activity; interruption-based capture fails. <https://andre-meyer.ch/the-work-life-of-developers-activities-switches-and-perceived-productivity/>
- C17** · WakaTime: voluntary always-on editor telemetry at scale when it returns personal value. <https://wakatime.com/time-tracking-for-developers>
- C18** · Nygard, Architecture Decision Records (2011): the canonical lightweight rationale template. <https://www.cognitect.com/blog/2011/11/15/documenting-architecture-decisions>
- C19** · ADRs in Practice, ECSA 2024 action research: friction is deciding what to document, not writing it. https://rebekkaa.github.io/files/2024_ECSA.pdf
- C20** · LLM-generated ADRs (context-strategy evaluation): the debrief agent's academic precedent. <https://arxiv.org/html/2604.03826v1>

I · Incentives and adoption

- I1** · Kankanhalli, Tan, Wei (MIS Quarterly 2005): intrinsic motivators dominate; feared loss of knowledge-power empirically weak. <https://misq.umn.edu/contributing-knowledge-to-electronic-knowledge-repositories-an-empirical-investigation.html>
- I2** · Cabrera and Cabrera (Organization Studies 2002): sharing as a public-goods dilemma; fix payoffs, efficacy, identity. <https://journals.sagepub.com/doi/10.1177/0170840602235001>
- I3** · Cerne et al. (Academy of Management Journal 2014): knowledge hiding triggers distrust loops that reduce the hider's own creativity. <https://journals.aom.org/doi/10.5465/amj.2012.0122>
- I6** · Rubinstein, "Alpha Capture": TOPS paid by idea performance, >\$250M/yr by 2004-05, ~\$30B AUM today. <https://www.netinterest.co/p/alpha-capture>

- I7 · Alpha capture systems overview: ideas carry rationale, timeframe, conviction; performance is quantified per contributor. https://en.wikipedia.org/wiki/Alpha_capture_system
- I8 · WorldQuant BRAIN consultant program: paid per alpha quality; top tiers \$8,000+ quarterly. <https://worldquantbrain.com/consultant>
- I9 · HBS case 413-702, Bridgewater: total decision capture (videotaped meetings) at the world's largest hedge fund. <https://www.hbs.edu/faculty/Pages/item.aspx?num=44831>
- I10 · Bridgewater transparency backfire coverage: published rankings produced fear and churn; capture decisions, not people-rankings. <https://factually.co/fact-checks/business/bridgewater-radical-transparency-controversies-governance-61a95e>
- I11 · Bornfeld and Rafaeli (First Monday 2017): badge launches raised contribution, median +88.6%, strongest for committed users. <https://firstmonday.org/ojs/index.php/fm/article/download/7299/6301>
- I12 · Yanovsky et al. (JASIST 2021): gamified credit measurably steers effort; aim it at debrief quality, not volume. Abstract-level. <https://arxiv.org/pdf/2008.06125>
- I13 · Marlow and Dabbish (CSCW 2013): employers trust third-party-verified contribution traces; "GitHub is your new resume." <https://www.cs.cmu.edu/~xia/resources/Documents/Marlow-cscw13.pdf>
- I14 · AIMA Due Diligence Questionnaires: allocators' standard forms demand succession plans, deputies, turnover tables. <https://www.aima.org/sound-practices/due-diligence-questionnaires.html>
- I17 · FCA SYSC 25.9, handover procedures: material "should include judgement and opinion, not just facts and figures." <https://handbook.fca.org.uk/handbook/SYSC/25/9.html>
- I18 · FINRA Notice 08-18 and NY DFS letters: mandatory ~2-week block leave with access cut; finance's compulsory bus-factor test. <https://www.finra.org/rules-guidance/notices/08-18>
- I19 · Jabrayilzade et al., "Bus Factor In Practice" (ICSE-SEIP 2022): 75% rate it important, 19% saw it measured; six person-months lost to one departure. <https://arxiv.org/pdf/2202.01523>

G · AI frontier and risk quantification

- G1 · Edge et al., Microsoft GraphRAG: entity graph + community summaries for global questions over a corpus; open source. <https://arxiv.org/abs/2404.16130>
- G2 · SWE-QA: repository-level code QA benchmark (intention understanding, cross-file reasoning). <https://arxiv.org/abs/2509.14635>
- G3 · CoMRAT: detects decision and rationale sentences in commit messages. <https://arxiv.org/html/2506.10986>
- G4 · Design-rationale mining as an established field (text-mining rationale; ASE 2022 rationale management). <https://dl.acm.org/doi/10.1145/3551349.3559568>
- G6 · Park et al., "Generative Agent Simulations of 1,000 People": two-hour interviews; ~85% of subjects' own test-retest consistency. <https://arxiv.org/abs/2411.10109>
- G7 · Rajaraman et al. (NeurIPS 2020): fundamental limits of imitation learning; clone error compounds off-distribution. <https://par.nsf.gov/servlets/purl/10237319>
- G8 · FlowHFT: imitation learning of trading policies; regime fragility is the known weakness. <https://arxiv.org/abs/2505.05784>
- G9 · Delphi ("digital minds"): shipping expert-clone product. <https://www.fastcompany.com/91356476/delphi-ai-digital-mind>
- G10 · The knowledge acquisition bottleneck (Expert Systems, 1988; Cooke 1994): the named 40-year-old problem. <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0394.1988.tb00065.x>
- G11 · LLMREI: LLM elicitation interviewer matches human interviewer error rates. <https://arxiv.org/abs/2507.02564>
- G12 · AutoElicit (ICML 2025): elicited knowledge as Bayesian priors; fewer labels, lower error. Demonstrated outside finance. <https://arxiv.org/abs/2411.17284>
- G13 · Settles and Meeder (ACL 2016): half-life regression for spaced repetition, deployed at Duolingo. <https://aclanthology.org/P16-1174/>
- G14 · Morgan Stanley x OpenAI: assistant used by >98% of advisor teams; "Debrief" meeting agent shipped. <https://openai.com/index/morgan-stanley/>
- G15 · BloombergGPT: in-house finance LLM on proprietary data. <https://arxiv.org/abs/2303.17564>
- G16 · JPMorgan LLM Suite: internal LLM portal for ~140-200K employees. <https://www.cnbc.com/2024/08/09/jpmorgan-chase-ai-artificial-intelligence-assistant-chatgpt-openai.html>

- G17** · RFC 3161 trusted timestamping; ANSI X9.95 financial-grade extension. <https://datatracker.ietf.org/doc/html/rfc3161>
- G18** · Tamper-evident Merkle logs (Crosby-Wallach; Google Trillian / Certificate Transparency infrastructure). <https://transparency.dev/verifiable-data-structures/>
- G19** · OpenTimestamps git integration: free cryptographic anchoring of commit hashes. <https://github.com/opentimestamps/opentimestamps-client/blob/master/doc/git-integration.md>
- G20** · Instructional fingerprinting of LLMs (NAACL 2024); provable model ownership after fine-tuning. <https://arxiv.org/abs/2401.12255>
- G21** · Avelino et al.: truck factor computed from git history; 46% of popular projects at TF=1; open-source tool. <https://peerj.com/preprints/1233/>
- G22** · Nassif and Robillard (ICSME 2017): turnover-induced knowledge loss as rare-severe-event risk. <https://www.cs.mcgill.ca/~martin/papers/icsme2017.pdf>
- G23** · Bird et al. (FSE 2011, Microsoft): ownership concentration predicts defects. <https://cabird.com/pdfs/bird2011dtm.pdf>
- G24** · Mockus and Herbsleb (ICSE 2002): Expertise Browser, who-knows-what from change history. <https://herbsleb.org/web-pubs/pdfs/mockus-expertise-2002.pdf>

L · Legal and regulatory

- L1** · Winston and Strawn, reasonable-measures primer: ~11% of trade-secret cases dismissed on reasonable measures; case list. <https://www.winston.com/en/insights-news/reasonable-measures-for-protecting-trade-secrets-a-primer>
- L2** · IPWatchdog (2026), reasonable particularity under the DTSA: claims dismissed for vague identification "regardless of whether actual misappropriation occurred" (NEXT v. CLEAResult; Syntel v. TriZetto won on documentation). <https://ipwatchdog.com/2026/01/28/identifying-trade-secrets-under-dtsa-critical-requirement-reasonable-particularity/>
- L4** · Asia IP on Jane Street v. Millennium: attorney prescription verbatim, "document development and implementation meticulously." <https://asiaiplaw.com/section/in-depth/can-a-trading-strategy-be-protected-as-a-trade-secret>
- L5** · DOJ SDNY (2025): trader arrested for stealing a quant firm's "alphas" developed at a cost exceeding \$1B; theft is now criminal enforcement. <https://www.justice.gov/usao-sdny/pr/trader-arrested-stealing-trade-secrets-global-quantitative-trading-firm>
- L6** · MiFID II RTS 6 (EU 2017/589), Art. 5(7): record when, who made, who approved, and the nature of every material algo change; Art. 9: annual self-assessment and validation report. <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32017R0589>
- L7** · FCA 2018 algo review: key decisions taken "without an adequate audit trail and without sufficient explanation of the underlying rationale." <https://www.fca.org.uk/publication/multi-firm-reviews/algorithmic-trading-compliance-wholesale-markets.pdf>
- L9** · Fed SR 11-7: documentation detailed enough for "parties unfamiliar with a model to understand how the model operates, as well as its limitations and key assumptions." <https://www.federalreserve.gov/supervisionreg/srletters/sr1107.htm>
- L10** · SEC Advisers Act Rule 204-2: written communications relating to recommendations and advice are required records. <https://www.ecfr.gov/current/title-17/chapter-II/part-275/section-275.204-2>
- L12** · Holland and Knight: \$3B+ in penalties, 100+ firms, for failing to preserve business communications since 2021. <https://www.hklaw.com/en/insights/publications/2024/12/a-long-winters-nap-sec-off-channel-communications>
- L13** · FCA SYSC 10A: mandatory recording including intent-stage communications; five-to-seven-year retention. <https://handbook.fca.org.uk/handbook/sysc10a>
- L14** · ICO monitoring-workers guidance (2023): lawful basis, proportionality, transparency, DPIA; the design checklist. <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/employment/monitoring-workers/>
- L15** · German BetrVG 87(1)(6): works-council co-determination over monitoring technology; architect as work-product capture. <https://www.employee-monitoring.net/compliance/employee-monitoring-laws-germany>
- L16** · ILPA Principles 3.0: key-person event should trigger automatic suspension of the investment period; succession planning demanded. https://ilpa.org/wp-content/uploads/2019/06/ILPA-Principles-3.0_2019.pdf
- L17** · Capco study (via ODD literature): ~50% of hedge fund failures are operational, vs 38% investment. [https://en.wikipedia.org/wiki/Operational_due_diligence_\(alternative_investments\)](https://en.wikipedia.org/wiki/Operational_due_diligence_(alternative_investments))

- L18** · Buy-over-build at the top: Global Relay serves 22 of the top 25 banks; Smarsh 18 of the top 20. <https://www.businesswire.com/news/home/20250113298920/en/Global-Relay-placed-in-2025-Gartner-Magic-Quadrant-for-Digital-Communications-Governance-and-Archiving-Solutions>
- L19** · SEC Rule 17a-4(f): tamper-evident storage plus a designated third party able to deliver records to regulators. <https://www.sec.gov/rules-regulations/staff-guidance/trading-markets-frequently-asked-questions/rule-amendments-broker>
- L20** · Federal Rule of Evidence 803(6): contemporaneous, regular-practice business records are admissible; the evidentiary payoff of ambient capture. https://www.law.cornell.edu/rules/fre/rule_803

Integrity notes: the Stanford twin result (G6) is on social-survey responses, not trading decisions; always present it with the imitation-learning limit (G7). The claim that pod shops make documentation a condition of capital allocation could NOT be verified publicly; do not assert it, use the risk-gating analogy plus SYSC 25.9 and AIMA DDQs instead. Bridgewater material (I9, I10) is a cultural precedent and a cautionary tale in one; cite both halves. Paywalled or abstract-only entries: I12, G10, and the HBS case (I9).